Curriculum Vitae (Min Seong Jo)

1. Contact Information

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2. Education

- **Chungnam National University (CNU)**, Daejeon, Republic of Korea
  -Ph.D. Program in Molecular Biotechnology, 2023 – Present
  -Research focus on Immunology with *in vitro-in vivo* correlation
- **Sungkyunkwan University (SKKU)**, Suwon, Republic of Korea
  -Master of Science in Biological Sciences, 2019 – 2021
  -GPA: 4.3/4.5
  -Master's Thesis: Genetic diversity and differentiation of Styrax obassia (*Styracaceae*) in Korea using microsatellite (SSR) markers.
- **Catholic University of Korea**, Bucheon, Republic of Korea
  -Bachelor of Science in Life Science, 2012 – 2019
  -GPA: 3.5/4.5
  -Undergraduate's thesis: Flora and Forest Community Structure of Wonmisan Mountain in Bucheon City, Korea (Korean)

3. Research Interest

A researcher specializing in transcriptomics, functional validation, and humanized mouse model development to enhance the precision of biopharmaceutical evaluation through *in vitro-in vivo* correlation.

Keywords: *Multi-omics Data Integration, In vitro-In vivo Correlation (IVIVC), Biopharmaceutical Efficacy & Safety Assessment, Precision Medicine & Biomarker Discovery*

4. Publications

1. Bak, I., Choi, M., Yu, E., Yoo, K. W., Jeong, S. Y., Lee, J., **Jo, M**., Moon, K. S., & Yu, D. Y. (2024). The Effects of Busulfan on Xenogeneic Transplantation of Human Peripheral Blood Mononuclear Cells in Recipient Mice. *Transplantation Proceedings*, *56*(2), 440–447. https://doi.org/10.1016/j.transproceed.2023.12.018

2. Jeong, S. Y., Park, D., Park, T., Han, J. S., Lee, J., Choi, C. H., **Jo, M**., Lee, Y. Bin, Kyun, M., Lang, Choi, M., Park, D., & Moon, K. S. (2024). Interspecies transcriptome profiles of human T cell activation and liver inflammation in a xenogeneic graft-versus-host disease model. *Heliyon*, *10*(23). https://doi.org/10.1016/j.heliyon.2024.e40559

3. **Jo, M**., Jeong, S. Y., Kyun, M., Lang, Lee, Y. S., Lee, J., Choi, C. H., Lee, Y. Bin, Choi, M., Rho, J., & Moon, K. S. (2026). Two-Step Algorithmic Selection of Interspecies Sequence-Mismatch-Based Housekeeping Genes for Precise Gene-Expression Assessment in PBMC-Humanized NSG Mice. *ACS Omega*, *11*(6), 9187–9200. https://doi.org/10.1021/acsomega.5c08313 **(Supplementary Cover)**

4. Kim, S. H., Kim, H. B., **Cho, M**. S., Kim, C. S., & Kim, S. C. (2020). Development and characterization of 17 microsatellite markers for Sonchus oleraceus. *Applications in Plant Sciences*, *8*(3), e11329. https://doi.org/10.1002/aps3.11329

5. Kyun, M., Lang, Kwon, J. I., Kim, J., **Jo, M**., Jung, H., Lee, Y. Bin, & Moon, K. S. (2025). Developing a Human Immune-Endothelial Coculture Model for Inflammatory Vasculitis and Drug Screening. *ACS Omega*, *10*(40), 47544–47558. https://doi.org/10.1021/acsomega.5c07484

6. Kyun, M. lang, Park, D., Jung, H., Ryu, J. H., Kim, I., Jeong, S. Y., Kim, J., **Jo, M**., Kwon, J. I., Kim, J. K., Kim, D., Park, S. A., Ryu, C. S., Kim, S. K., Lee, Y. Bin, Park, D., & Moon, K. S. (2025). 3D liver model for assessing immune-mediated hepatotoxicity from biopharmaceutical risks. *Chemical Engineering Journal*, *523*, 168419. https://doi.org/10.1016/j.cej.2025.168419

7. Choi, M., Park, D., Lee, D. H., Choi, C. H., **Jo, M**., Jeong, S. Y., ... & Moon, K. S. (2026). Wnt Signaling Downregulation Mediates T Cell Apoptosis Following OKT3-Induced T Cell Activation in Preclinical Models. The FASEB Journal, 40(6), e71700.

5. Research Experience

Graduate Researcher (Joint Program), Center for Global Biopharmaceutical Research Group, Korea Institute of Toxicology (KIT) & Chungnam National University (CNU)
2023 – Present

1) **In vitro-In vivo Correlation system**
   -Investigated the human immune system using NOG and NSG immunodeficient mice, focusing on the correlation between cytokine profiles and immune responses
   -Key Techniques: Flow cytometry, ELISA, RT-qPCR, RNA extraction, cDNA synthesis, in vivo and in vitro experiments (orbital blood collection, tail vein injection, autopsy)

2) **Monoclonal Antibody (mAb) Therapeutic Research**
   -Demonstrated the efficacy of Anti-CD3 in attenuating liver fibrosis by modulating T-cell and macrophage responses
   -Key Techniques: Monoclonal antibody handling/administration, T-cell isolation from hPBMC, histology, flow cytometry, macrophage polarization analysis, histological

analysis of liver tissue, hydroxyproline assay

3) **Development of Species-Specific qPCR Markers for Humanized Mice (Jo et al., 2026)**
   -Published a two-step algorithmic method to select housekeeping genes based on interspecies sequence mismatches (Human vs. Mouse)
   -Key Techniques: Bioinformatics-based sequence alignment, Interspecies primer design (Mismatch-based), qPCR validation

4) **Disease Model Establishment with Humanized Mice (Bak et al., 2024; Jeong et al., 2024)**
   -Established disease and Graft-versus-Host Disease (GvHD) models in NOG mice to study the correlation between cytokines and resistant cell populations in the context of the human immune system
   -Key Techniques: Bulk and single-cell RNA-seq, RT-qPCR, RNA extraction, cDNA synthesis, flow cytometry, ddPCR, in vivo experiments

5) **In vitro Disease Modeling (Kyun, Kwon, et al., 2025; Kyun, Park, et al., 2025)**
   -Developed a human immune-endothelial coculture model to simulate inflammatory vasculitis for pathophysiology studies and high-throughput drug screening
   -Established a 3D liver model to assess immune-mediated hepatotoxicity associated with biopharmaceutical risks
   -Key Techniques: 3D cell culture, Hepatocyte-Immune cell co-culture, Cytotoxicity assays, Live/Dead staining, Confocal microscopy, Multiplex cytokine profiling

Internship, Advanced Biopharmaceutical Research Group, Korea Institute of Toxicology (KIT)
2022 – 2023

1) **Mimic the Human Immune System with Humanized Mice Model**
   -Investigated the human immune system using NOG and NSG immunodeficient mice, focusing on the correlation between cytokines and immune responses
   -Key Techniques: Flow cytometry, ELISA, RT-qPCR, RNA extraction, cDNA synthesis, in vivo experiments (orbital blood collection, tail vein injection, autopsy)

Researcher, Genome Editing Laboratory, School of Medicine, SKKU

2021 – 2022

1) Research on the development of next-generation genome editors
-Developing next-generation genome editors, such as Prime and Base editors, uses proteins with potential applications based on dCas9 and nCas9
-Key Techniques: Vector cloning, colony PCR, in vitro transcription, in vitro deamination, protein overexpression, protein purification, SDS-PAGE, electroporation, transfection (lipofectamine), NGS(deep sequence), Sanger sequence, gDNA extraction, gel electrophoresis, etc

2) Point mutation-based EGFR T790M drug development research
-Participated in drug development research with a genome editing study to correct

T790M in EGFR, a side effect of drugs used to treat sepsis patients
-Key Techniques: Cell work (cell line: H1299, PC-9), single cell library, drug test, NGS(deep sequence), Sanger sequence, gDNA extraction, PCR, electrophoresis, transfection (lipofectamine), sgRNA design, sgRNA extension, in vitro transcription, RNA purification, oligo design, etc

Graduate Student(Full-time), Plant Systematics Laboratory, Department of Biological Sciences, SKKU
2019 - 2021

1) Population genetics study of *Styrax obassia*
-Conducted a population genetics study using 14 SSR markers on 288 accessions of *Styrax obassia* across the Korean Peninsula
-Performed distribution modeling using machine learning
-Key Techniques: cpDNA extraction, PCR, DNA sequencer, GenALEx v. 6.5, GraphPad Prism v. 8.0, POPULATIONS v.1.2.30, STRUCTURE v.2.3.4, STRUCTURE HARVESTER v.0.6.94, CLUMPAK, CLUMPP, Bottleneck v.1.2.02, Maxent 3.4.1, GBIF, WorldClim 2, Linux, R studio, Python, etc
-Thesis: Link to Master's thesis

2) Development of 17 microsatellite markers in the *Sonchus oleraceus* with population genetics (Kim et al., 2020)
-Analyzed WGS based on cpDNA from Sonchus oleraceus and four other species collected in Korea and Spain
-Developed 17 microsatellite markers based on the analysis for population genetics studies
-Responsibilities: Managed overall experiments, including cpDNA extraction, NGS data handling (Linux, Python), NGS data multiple assembly (Linux), SSR marker development, PCR, and DNA sequencing

Teaching Assistant, 2019 - 2020 Department of Biological Sciences, SKKU Experimental of General Biology, Laboratory of Plant Taxonomy

Undergraduate Student, 2017 - 2019 Ecology Laboratory, Department of Life Sciences, Catholic University of Korea, Community Ecology, Plant Ecology

5. Awards

1) Outstanding Researcher Award in the KIT, 2026, President of the Korea Institute of Toxicology
2) Outstanding Achievement Award in the Research Achievement Competition, 2024, President of the Korea Institute of Toxicology
3) Outstanding Poster Presentation Award in the KIT annual conference, 2023, President of the Korea Institute of Toxicology

4) Seoul City Chairman's Citation, 2023, Chairman of Seoul
5) Awarded as an excellent eco-activist group, 2018, President of the National Institute of Biological Resources

6. References

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I declare that the details provided above are true.   (Minseong Jo) 조민성